WASTEWISE-ML · USER REQUIREMENTS

WasteWise-ml user requirements

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DOCUMENT CONTROL

Document ID	WML-URS-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/03a-user-requirements.md @ 9031e25		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First user requirements (kit 1.8 upgrade), drawn from the users and stakeholders in WML-PRB-001 v0.5

WasteWise-ml user requirements

What users need, written from their side. Product requirements that meet these needs are in WML-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix WML-RTM-001 (`03b-traceability.md`). The needs are drawn from the users and stakeholders in WML-PRB-001 Tables 1 and 2; none has yet been confirmed in a co-design session with the first candidate partner (SWaCH, Pune, decided 2026-10-02), so every priority is a proposal. Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Right material class	When I show an item, I am told its material class correctly almost every time it gives an answer	High	Informal waste picker; cooperative operator (PRB Table 1)
UR-002	Performance	Buyer's grade	For plastics and other graded material, I am told the grade my buyer pays for, where a photo can tell it	High	Waste picker; scrap buyer (PRB Table 1)
UR-003	Safety	Hazards flagged	Batteries, sharps and chemical containers are flagged as hazards and never offered as something to sort, and the station does not add a fire or cut risk	High	Waste picker; cooperative; municipality (PRB Tables 1 and 2)
UR-004	Performance	Honest unsure	When it is not sure it says so rather than guessing, and it still answers most items	High	Waste picker; scrap buyer (PRB Table 1)
UR-005	Performance	Fast	The answer comes quickly enough not to slow work that is paid by weight	High	Waste picker (PRB operating environment)
UR-006	Usability	No reading needed	I can use it without reading, in my own language, after a short briefing, and see or hear the answer from where I stand	High	Waste picker; cooperative (PRB Table 1, literacy)
UR-007	Usability	Hands free, no stretching	My hands stay free, the item lies still, and I can trigger a scan without reaching up or leaning in	High	Waste picker in gloves (PRB operating environment)
UR-008	Logistics	My own phone, offline	It runs on a phone I own or one costing about USD 150, with no network and a small download	High	Waste picker; cooperative (PRB constraints)
UR-009	Performance	Lasts a shift	The phone lasts a full shift of scanning	Medium	Waste picker (PRB operating environment)
UR-010	Usability	Our grade names	Grade names and bin numbers match our buyer's price list, and the cooperative can change them	High	Cooperative; scrap buyer (PRB Table 1)
UR-011	Regulatory	Our data, our consent	Photos are taken only with consent, show no faces, are uploaded only if we agree, and the field data is co-owned by our organization	High	Cooperative; waste pickers (PRB constraints, D6)
UR-012	Regulatory	Open and licensed	Code, model weights and dataset licenses are open and checked, so others can inspect and extend them	Medium	Researchers and NGOs (PRB Table 1)
UR-013	Performance	Proven on real waste	The accuracy is shown on real field photos from our kind of site before we rely on it	Medium	Scrap buyer; researchers; cooperative (PRB Table 2)
UR-014	Deployment and assembly	Build it locally		Medium	Cooperative operator (PRB Table 1)

ID	Category	Title	Need	Priority	Source
			The station can be built from local parts with hand tools and sits on a bench we already have		
UR-015	Environment and durability	Works in our shed	The station copes with wet and dirty items, heat and dust in a sorting shed	Medium	Cooperative operator (PRB operating environment)
UR-016	Cost	Low cost	The station costs little enough for a cooperative to buy or build	Medium	Cooperative; NGOs (PRB constraints)
UR-017	Maintenance	Easy to keep going	The phone lifts out to charge or swap, and parts can be replaced locally	Low	Cooperative operator (PRB Table 1)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.